
MIDTERM EXAM 3 (FORM B)

NAME: _____

ID N°: _____

INSTRUCTIONS

- This is a closed-book exam. You may not use your notes, textbook or any outside resources to assist you.
- Write your answer to each question in the space assigned to each problem. If you want the instructor to read the scratch working page for the answer to a specific problem, explicitly write that on the space assigned to said problem.
- Always remember to show your work. An answer without a proper justification will earn little to no credit, even if it is correct.
- Circle or box your final answer to each question.
- Make sure that your answers are organized and legible. Use a dark pencil or a pen. Any work that is hard or impossible to read will not be graded.
- When solving a problem, if you are going to use an alternative method to the ones discussed in class (for instance, inverting a matrix via adjugates or using Cramer's rule to solve a linear system), you are required to include a short explanation on how this alternative method works.
- The use of calculators of any type is not allowed.

This exam consists of four problems, and your total score will be out of $10 + 10 + 10 + 10 = 40$ points.

Good Luck!

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Problem 1 (10 points). Answer the following questions concerning linear independence.

- (a) (5 points) Let V be a vector space and suppose $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is a set of vectors in V . State precisely what it means for the set $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ to be a linearly independent set in V .
- (b) (5 points) Suppose $\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ is a linearly independent set in V . Determine if $\{\mathbf{x}, \mathbf{y} - \mathbf{x}, \mathbf{z}\}$ is a linearly independent set.

Problem 2 (10 points). Let $\mathbf{P}_3$ be the vector space of all polynomials of degree at most 3, including the zero polynomial. We define a transformation $T: \mathbf{P}_3 \rightarrow \mathbb{R}^2$ via the equation

$$T(p(x)) = (p(0), p(2)), \quad \text{for all } p(x) \in \mathbf{P}_3.$$

- (a) (4 points) Show that T is a linear transformation. (The question continues on the next page)

- (b) (6 points) Compute a basis and the dimension of the kernel $\ker T$.

Problem 3 (10 points). Consider the matrix

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

The characteristic polynomial of A is $c_A(x) = (x - 1)(x - 3)^2$.

(a) (2 points) The matrix A has two eigenvalues λ_1 and λ_2 with $\lambda_1 > \lambda_2$. What are their values?

(b) (4 points) Find a set of basic eigenvectors for A associated with the eigenvalue λ_1 .
(The question continues on the next page)

- (c) (4 points) You are given that the vector $\mathbf{z} = (0, 1, -1)$ is an eigenvector of A with eigenvalue λ_2 . Is A diagonalizable? If so, find a diagonal matrix D and an invertible matrix P such that $D = P^{-1}AP$. Otherwise, explain why A is not diagonalizable.

Problem 4 (10 points). Let

$$B = \begin{bmatrix} 1 & 0 & 2 & 0 \\ -2 & 2 & -6 & 14 \\ -3 & 5 & -11 & 35 \end{bmatrix}, \text{ whose reduced row-echelon form is } R = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 7 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

(a) (5 points) Find a basis for the row space $\text{row } B$ and a basis for the column space $\text{col } B$.

(b) (5 points) State precisely what is the null space of B , $\text{null } B$. Then, compute the dimension of $\text{null } B$.

(**Hint:** you are not asked to find $\text{null } B$ explicitly).

Scratch working page

Scratch working page